AI 100 Final Project Report

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1. What I Learned About GenAI

Through this project, I learned that GenAI is not a tool that directly gives solutions, but rather a system that helps guide thinking. In this assignment, GenAI was used to evaluate my self-reflections and determine whether my reasoning was deep or superficial. One important thing I noticed is that GenAI often labeled my initial reflections as “Bad” when they were too vague or only described what happened instead of why it happened. For example, when I removed feature scaling, my initial reflection was that the model performed worse. However, GenAI guided me to think deeper by asking how feature magnitude affects gradient updates. This helped me understand that larger feature values dominate the learning process and cause unstable training.

Another example was when I used the full dataset for both training and testing. Initially, I only noticed that the accuracy was very high. GenAI pushed me to think about why that might happen, which led me to recognize the issue of data leakage. This showed me that GenAI is useful for improving reasoning, not just answering questions. Overall, I learned that GenAI works best as a cognitive partner. It helps identify weaknesses in thinking and encourages deeper analysis through guided questions instead of direct answers.

2. What I Learned About Building an AI System

This project helped me understand that building an AI system is not just about writing code, but about carefully managing data, model design, and training setup. One key lesson is that small changes can have a large impact on performance. For example, removing scaling, changing the learning rate, or modifying the number of training epochs all significantly affected the model’s behavior. Some changes caused the model to fail completely, while others led to misleading results, such as extremely high accuracy due to data leakage. I also learned that many issues in AI systems are not obvious code errors but come from incorrect assumptions. For instance, changing the order of features in the test data caused the model to perform poorly, even though the code itself ran without errors. This showed me that consistency between training and testing data is critical.

Another important lesson is that debugging AI systems is an iterative process. I had to repeatedly test, reflect, and refine my understanding of what was going wrong. The combination of my own reasoning and GenAI’s guidance helped me identify the root causes of issues more effectively. Overall, I learned that building AI systems requires attention to detail, careful experimentation, and strong reasoning skills. It is not just about getting a working model but understanding why it works and how different components interact.